

12.6 Cylinders and Quadric Surfaces

Prerequisite information for Section : Question #1 below has *EXACTLY ONE* correct answer. Questions #2 - #5 each have *MORE THAN ONE* correct answer. For questions #2 - #5, please report one correct answer of your choosing. Also, for all those wonderful people who first met me in MAT 151 ... **now you know why we did this!**

1. Give the general (2-D) equation (standard form) for a circle with radius r centered at the point (h, k) .

$$(x-h)^2 + (y-k)^2 = r^2$$

2. Give the general (2-D) equation for an ellipse centered at the point (h, k) with major axis (in x or y -your choice) of TOTAL length 10 and minor axis of TOTAL length 6.

$$\frac{(x-h)^2}{100} + \frac{(y-k)^2}{36} = 1$$

3. Give an equation (in 2-D) for a parabola with vertex (h, k) which opens in the positive y direction.

$$x = (y-k)^2 + h$$

4. Give an equation (in 2-D) for a parabola with vertex (h, k) which opens in the negative x direction.

$$y = (x-h)^2 + k$$

5. Give an equation (in 2-D) for a hyperbola centered at $(0, 0)$.

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1, \quad a, b \in \mathbb{R}$$

Definitions in this section: Give the definition of each term below, using the given passage from your text/e-text.

- **Traces:** [first paragraph of the section; see the video for more information]

Curves of the intersections of a surface with the planes parallel to a coordinate plane.

- **Cylinder:** [not just a soup can! definition is above Example 1 of Section 12.6; see the video for more information]

A surface consisting of all lines parallel to a given line that pass through a plane curve.

- **Quadric Surface:** [definition is above Example 3 of Section 12.6]

A second degree equation in variables $x, y, z \in \mathbb{R}$

$$\left. \begin{array}{l} Ax^2 + By^2 + Cz^2 + J = 0 \\ Ax^2 + By^2 + Iz = 0 \end{array} \right\} \text{standard forms}$$

Full form w/ all translations,
/

$$Ax^2 + By^2 + Cz^2 + Dxy + Eyz + Fxz + Gx + Hy + Iz + J = 0$$

Topics to learn in this section:

1. Cylinders as three-dimensional extensions of curves in two variables.
2. Quadric surfaces as the graphs of second-degree polynomials in x , y , and z .
3. Using traces to visualize and draw graphs of functions.

ok

13 Vector Functions

13.1 Vector Functions and Space Curves

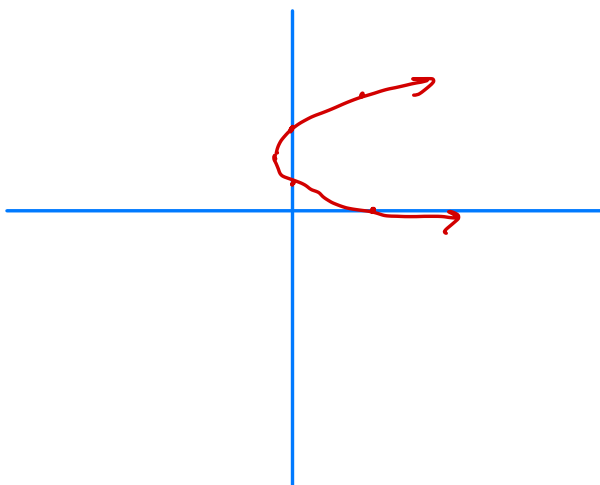
Prerequisite information for Section 13.1:

Carefully sketch and identify (name) the curve defined by the parametric equations,

$$\begin{aligned}
 x &= t^2 - 2t & \nearrow & \quad x = (y-1)^2 - 2y + 2 \\
 y &= t + 1 & \Rightarrow y-1 = t & \quad = y^2 - 2y + 1 - 2y + 2 \\
 & & & \quad = y^2 - 4y + 3
 \end{aligned}$$

where t is a real number. (For your sketch, I recommend choosing integer values for t between -3 and 3.) Then rewrite this description of the curve as one equation in x and y (no t). (Hint: Review Chapter 10 of your text.)

$$x = y^2 - 4y + 3$$



Definitions in this section: Give the definition of each term below, using the given passage from your text/e-text.

- **Vector-Valued Function, or Vector Function:** [discussion before Example 1 of Section 13.1]

A vector valued function has a domain of $\mathbb{R}^n$ and a range of V_n , where V_n denotes an arbitrary set of vectors

$$r: \mathbb{R}^n \rightarrow V_n$$

- **Component Functions:** [discussion before Example 1 of Section 13.1]

Component functions are functions f, g, h of reals where a vector function is defined as

$$\vec{r}(t) = f(t)\hat{i} + g(t)\hat{j} + h(t)\hat{k}, \quad t \in \mathbb{R}$$

- **Space Curve:** [discussion surrounding Box 2 and before Figure 1 of Section 13.1; the video is also related]

A space curve is the set (of points (x, y, z) given by $x = f(t), y = g(t), z = h(t), t \in I \subseteq \mathbb{R}$

Topics to learn in this section:

1. The connection between space curves and ranges of vector functions.
2. Matching vector equations with their curves.
3. Parametrizations of curves in space are not unique.
4. Visualization of curves in three dimensions.